

Step-by-Step Trace

Iter	Before Operation	left→value	right→value	Case	Action	After Operation
1	[12, -7, -5, 70, -3, 15] L=0, R=5	12	15	both ≥ 0	no swap	[12, -7, -5, 70, -3, 15]
2	[12, -7, -5, 70, -3, 15] L=1, R=4	-7	-3	$L < 0 \ \& \ R < 0$	no swap	[12, -7, -5, 70, -3, 15]
3	[12, -7, -5, 70, -3, 15] L=2, R=4	-5	-3	$L < 0 \ \& \ R < 0$	no swap	[12, -7, -5, 70, -3, 15]
4	[12, -7, -5, 70, -3, 15] L=3, R=4	70	-3	$L \geq 0 \ \& \ R < 0$	swap 70 ↔ -3	[12, -7, -5, -3, 70, 15]
5	L(4) > R(3)	–	–	termination	loop ends	final: [-7, -5, -3, 70, 12, 15]